Will It Rain Tomorrow?

PREDICTING RAIN IN AUSTRALIA WITH MACHINE LEARNING

A Python classification project: from 145,460 daily weather records to a model that warns about rain one day ahead.

Presented by Manar Ben Salem · Wiem Habbassi · Ghofrane Ayari

Records 145k
Places 49

Rain
next day

Share 22%
Hum.3pm 69%

Best
SVM model

AUC 0.86
Recall 77%

Dry
next day

Share 78%
Hum.3pm 47%

The Business Question



GIVEN TODAY'S WEATHER, WILL IT RAIN TOMORROW?

Farmers

plan irrigation and harvest

Event organisers

decide on cover or postponement

Construction

schedule outdoor work safely

Airports

prepare staff and equipment

Data Description





SOURCE & SIZE

- weatherAUS.csv: daily observations from Australian weather stations (Kaggle)
- 145,460 days × 23 columns
- 49 locations, Nov 2007 - Jun 2017

VARIABLES

- 16 numerical: temperature, rainfall, wind, humidity, pressure, cloud, sunshine
- 6 categorical: location, 3 wind directions, RainToday
- Target: **RainTomorrow** (Yes / No)

Descriptive Analysis



AVERAGES: DRY NEXT DAY vs RAINY NEXT DAY

Humidity 3pm

46.5% on dry days vs 68.8% when rain follows

Pressure 9am

1018.6 vs 1014.4 hPa - lower pressure before rain

Rainfall today

1.3 vs 6.1 mm - wet days tend to follow wet days

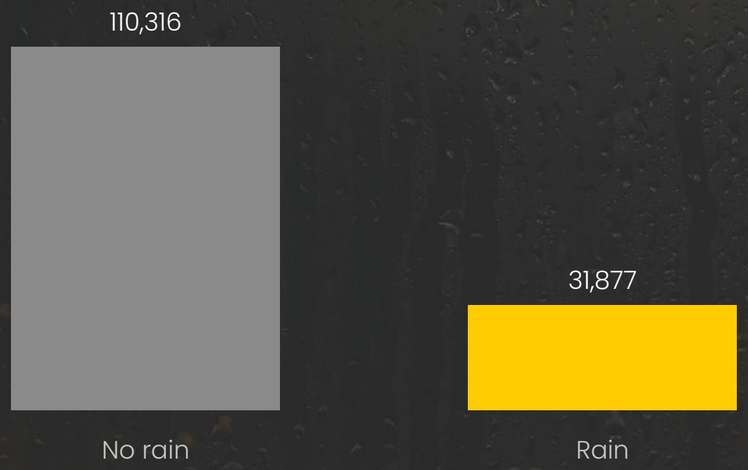
Wind gusts

38.3 vs 45.9 km/h - stronger gusts before rain

Rain Is Rare and Seasonal

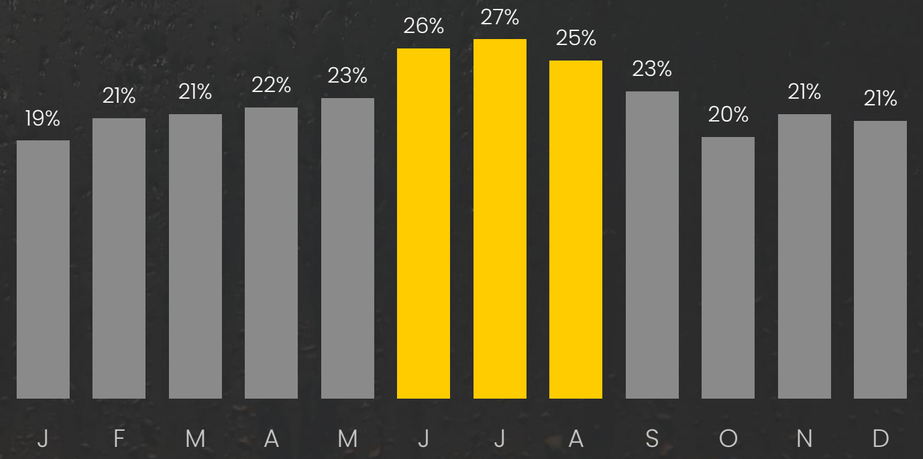
DATA VISUALIZATION

Rain tomorrow? (all 142,193 labelled days)



Only 22% of days are followed by rain → unbalanced target

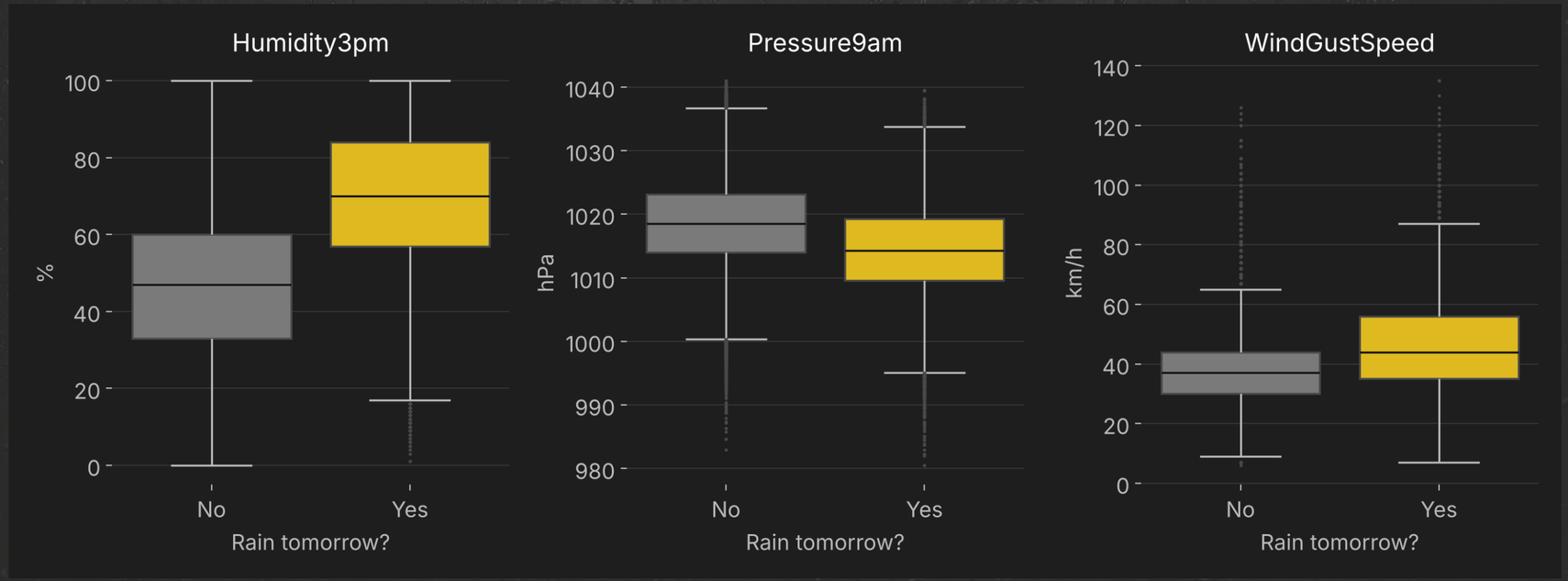
% of days followed by rain, by month



Winter (June-August) is the rainiest season

What Comes Before Rain

DATA VISUALIZATION · BOXPLOTS BY RAINTOMORROW

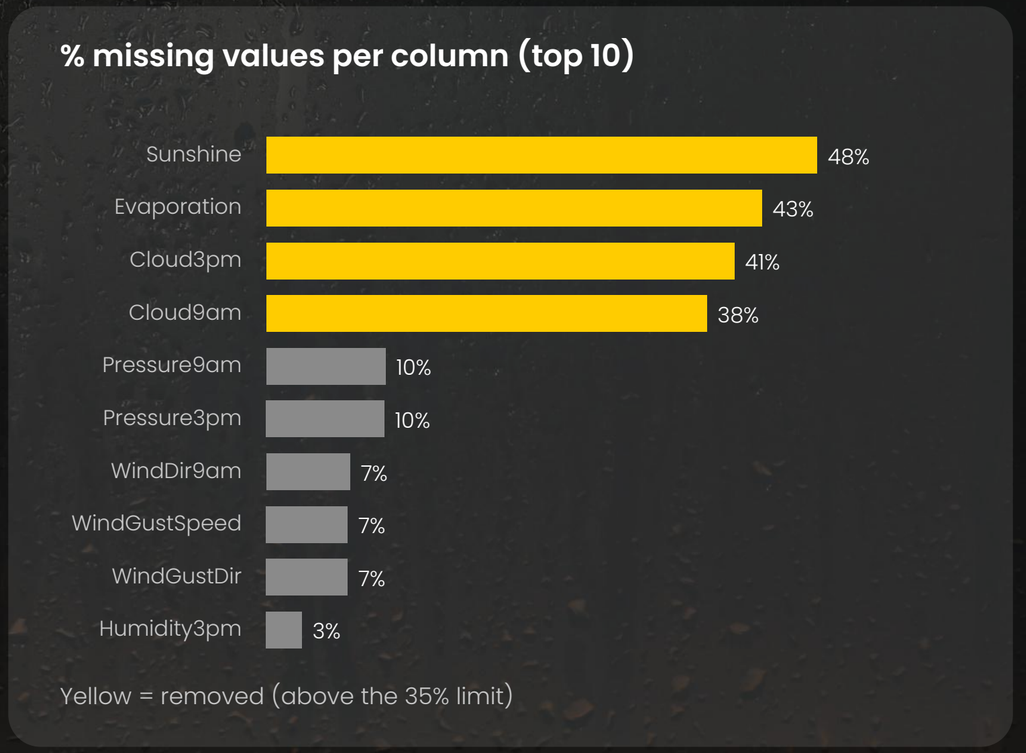


Before a rainy day: higher afternoon humidity, lower pressure and stronger wind gusts.

Data Preparation

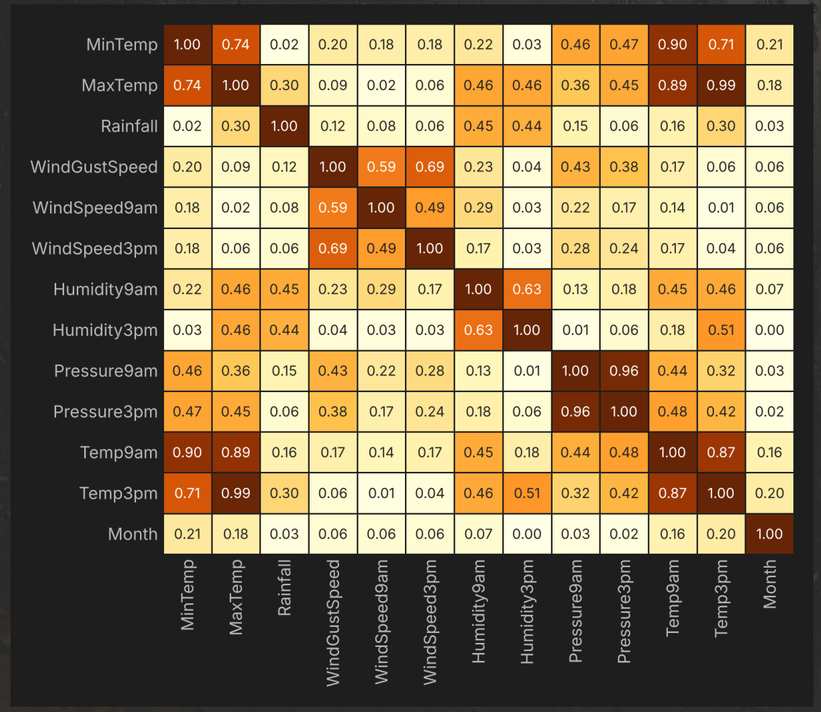
FROM RAW RECORDS TO A CLEAN MODELLING TABLE

1. Removed duplicates and the 3,267 rows with no RainTomorrow value
2. Dropped 4 columns with > 35% missing: Sunshine, Evaporation, Cloud9am, Cloud3pm
3. Date → Month; stratified random sample of 20,000 rows (as in the diamonds lab)
4. Filled missing values: median for numbers, mode for categories
5. RainToday & target → 0/1; Location and wind directions → dummy variables
6. Rare-category rule (< 30 rows → "Other"): not needed, smallest has 1,578



Removing Redundant Features

FEATURE SELECTION · STEP 1 · RELATIONSHIP > 0.80



0.97 Drop **Temp3pm** - keep **MaxTemp**

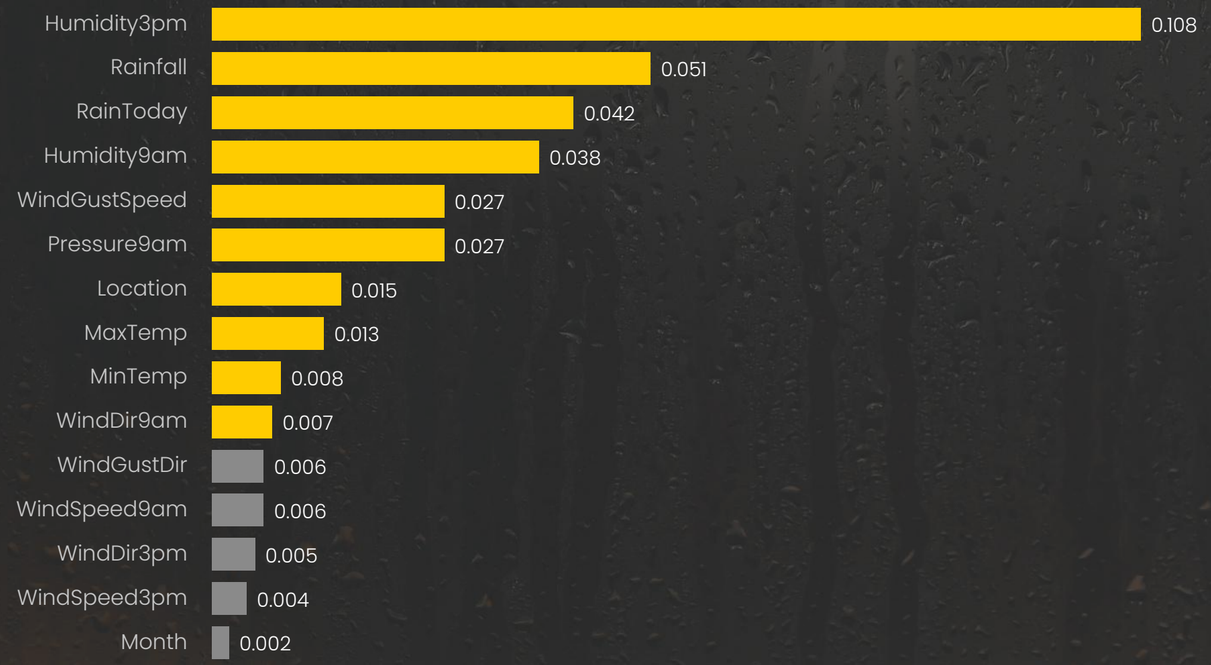
0.95 Drop **Pressure3pm** - keep **Pressure9am**

0.90 Drop **Temp9am** - keep **MinTemp**

Spearman for numbers, Cramér's V for categories. No categorical pair was above 0.80, so all wind directions were kept.

Keeping the 10 Most Informative

FEATURE SELECTION · STEP 2 · SELECTKBEST (MUTUAL INFORMATION)



K = 10

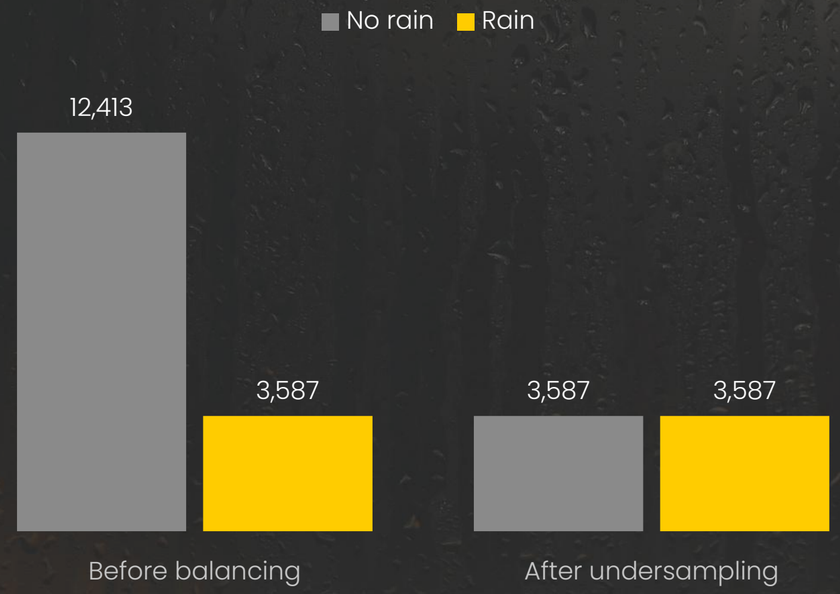
features kept (yellow)

- Humidity3pm carries twice the information of any other feature
- Dropped: WindGustDir, WindSpeed9am/3pm, WindDir3pm, Month
- After dummy encoding: 73 model inputs

Fixing the Unbalanced Target

RANDOM UNDERSAMPLING · TRAINING SET ONLY

Training set: rows per class



Why?

A model can reach 78% accuracy by always saying "no rain" - and miss every rainy day.

How?

Keep all 3,587 rainy days + a random 3,587 dry days → 50 / 50.

Why not oversampling?

Copied rows could land in both CV training and test folds; we have enough data to undersample.

Fair test

The 4,000 validation days keep the real 78 / 22 mix.

Model Building

5 CLASSIFIERS · 5-FOLD GRIDSEARCHCV · SCORED ON ROC AUC

Balanced training set
7,174 days
→
StandardScaler
(inside pipeline)
→
Classifier
→
GridSearchCV
5 folds
→
Validation set
4,000 days

Model	Hyperparameters searched	Best setting
Logistic Regression	C: 0.1, 1, 10	C = 1
SVM	C: 0.1, 1, 10 · gamma: scale, auto	C = 1, gamma = scale
Decision Tree	max_depth: 3, 5, None · min_samples_leaf: 1, 5	depth 5, leaf 5
Random Forest	n_estimators: 100, 200 · max_depth: 5, None	200 trees, no depth limit
Gradient Boosting	n_estimators: 100, 200 · learning_rate: 0.05, 0.1	200 trees, rate 0.1

Model Performance

VALIDATION SET · 4,000 DAYS WITH THE REAL 78 / 22 MIX

Model	CV AUC	Val. AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.840	0.853	78.0%	50.7%	74.0%	0.602
SVM	**0.848**	**0.860**	**79.1%**	**52.4%**	**77.0%**	**0.624**
Decision Tree	0.805	0.815	76.1%	47.8%	71.1%	0.571
Random Forest	0.846	0.858	78.5%	51.3%	76.0%	0.613
Gradient Boosting	0.849	0.853	78.0%	50.7%	74.1%	0.602

BEST MODEL

SVM

0.860

validation AUC (highest)

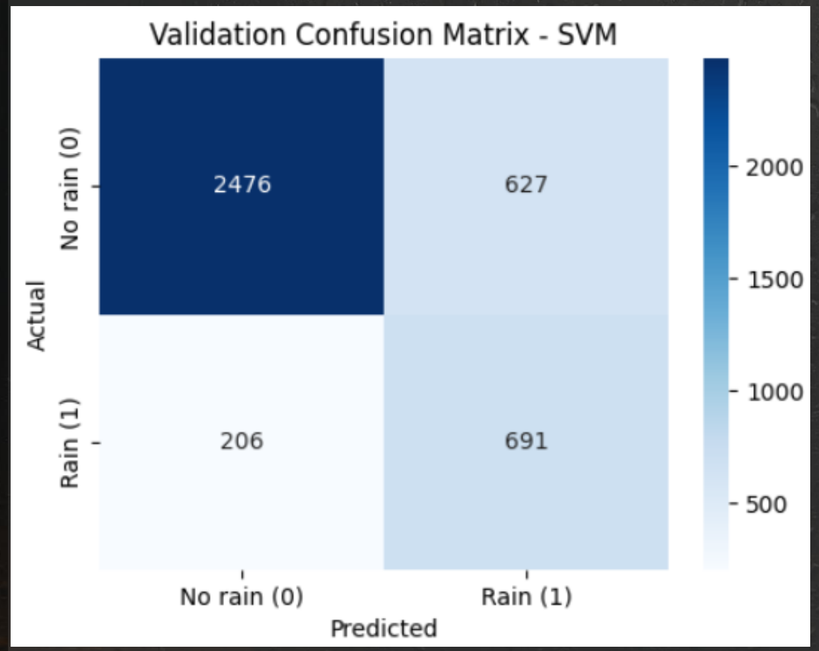
77%

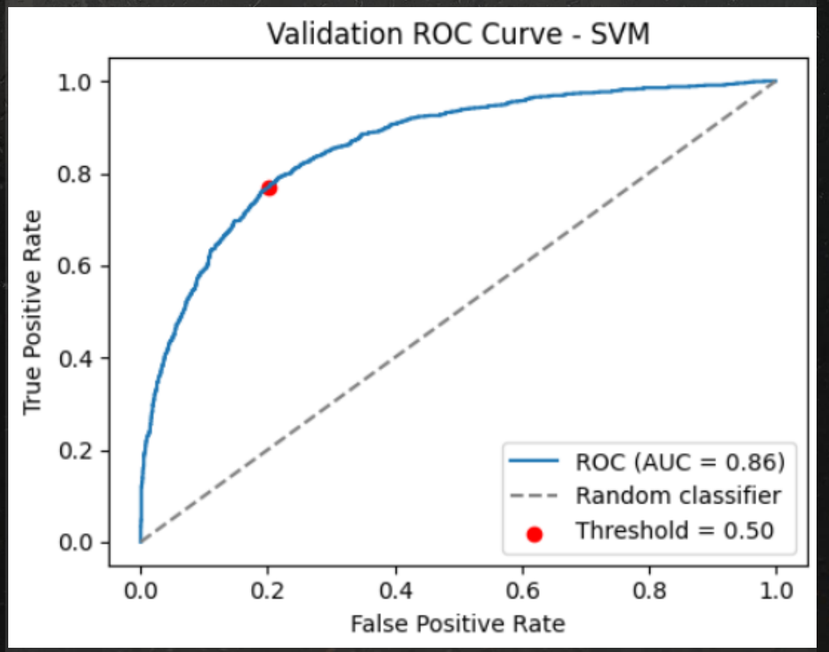
of rainy days caught

All five models are close; the single Decision Tree is weakest.

Inside the Best Model

SVM · CONFUSION MATRIX AND ROC CURVES (VALIDATION SET)





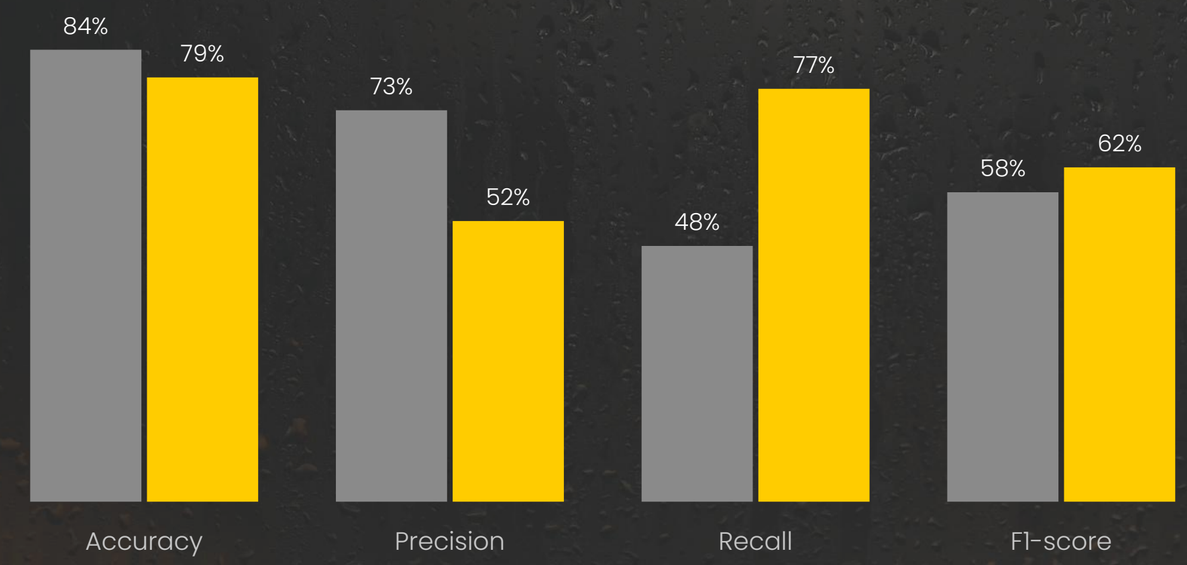
691 / 897
rainy days caught (recall 77%)

2,475
dry days correctly cleared

628
false alarms → precision 52%

Did Balancing Help?

SAME SVM, TRAINED WITH AND WITHOUT UNDERSAMPLING



RAINY DAYS CAUGHT

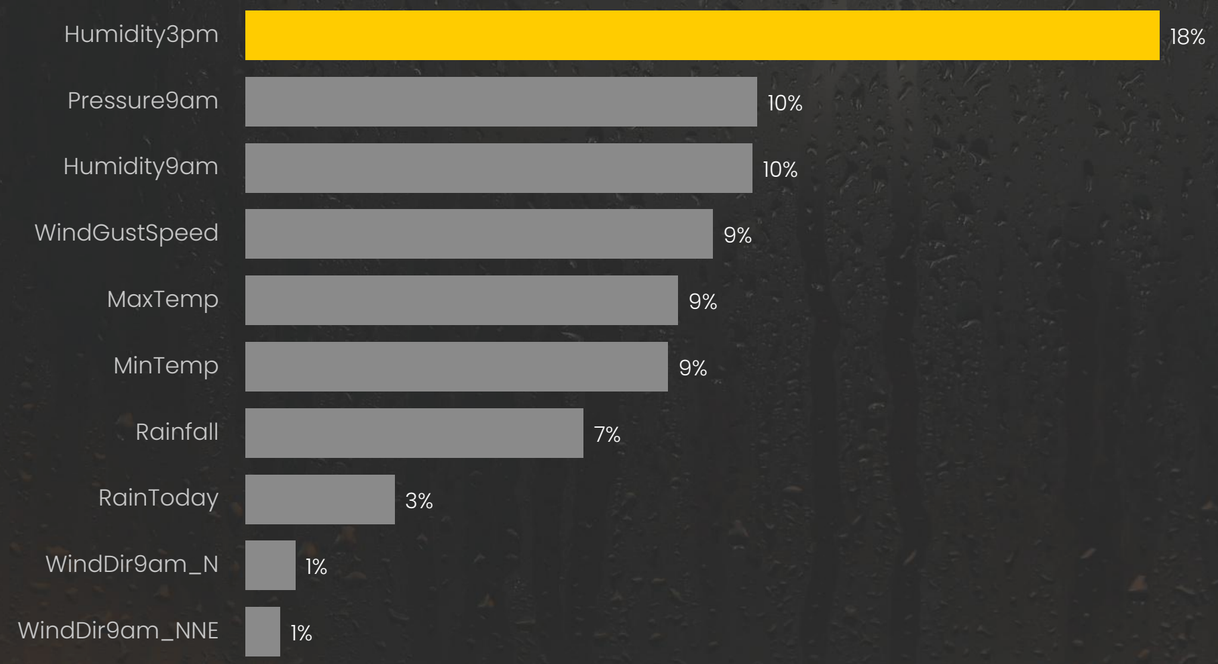
48% → 77%

Accuracy falls (84% → 79%) because the unbalanced model mostly says "no rain".

F1 improves: we trade some false alarms for many more rainy days caught - the right trade when missing rain is costly.

What Drives Tomorrow's Rain

RANDOM FOREST FEATURE IMPORTANCE (TOP 10)



Humidity at 3pm

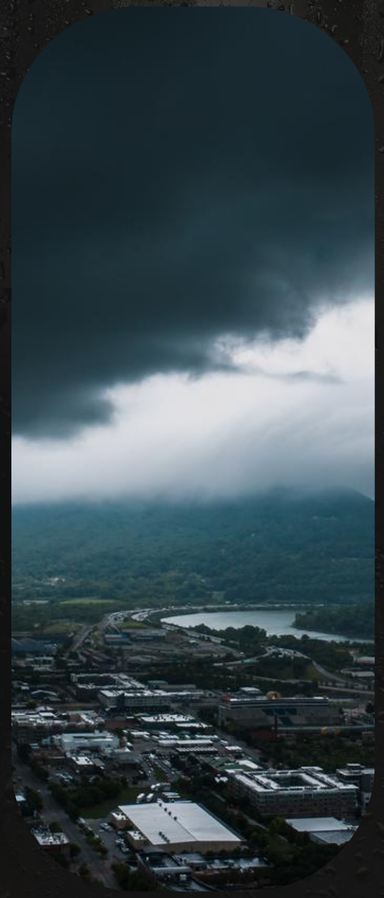
18% of importance - the #1 signal

Pressure & morning humidity

low pressure and moist air come before rain

Wind gusts & temperature

stronger gusts and cooler days point to rain


Decision Making

1. **Deploy SVM as a daily alert**

   Today's readings → a "rain tomorrow" warning.

2. **Tune the threshold per user**

   Lower it for farmers and airports who can't miss rain; raise it to cut false alarms.

3. **Watch humidity & pressure**

   High 3pm humidity with falling pressure is the simplest warning sign.

4. **Improve the data**

   Sunshine and cloud were 40-48% missing; collecting them could lift accuracy.

Thank You

QUESTIONS?

Will it rain tomorrow? With an SVM trained on balanced data, we catch 77% of rainy days (AUC 0.86).

Manar Ben Salem · Ghofrane Ayari · Wiem Habbassi